Analysing the Impact of **Gross Domestic Saving** on **Gross Domestic Product in India**

An annual time-series econometric analysis using data
Period Covered: 1990-2023 (Annual)

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Problem Statement and Economic Intuition

- **Gross Domestic product, or GDP**, This measures the total value of all finished goods and services produced within a country's borders in a specific time period. It represents the "Income" or "Output" of the nation. .
- A **Gross Domestic saving or GDS,**. This is the portion of the GDP the is not consumed. It is calculated as
- =GDP- Final consumption expenditure . It represent the pool of funds available for investment .

HYPOTHESIS:

Ho: GDSs do not significantly affect GDP growth.

H1: Higher GDSs significantly increase GDP growth.

Data Description

- **Data type:** Annual time-series data.
- **Time period:** 1990–2023.
- **Number of observations:** 34
- **Dependent variable Y:** GDP Growth Rate.
- **Independent variable X:** Gross Domestic saving

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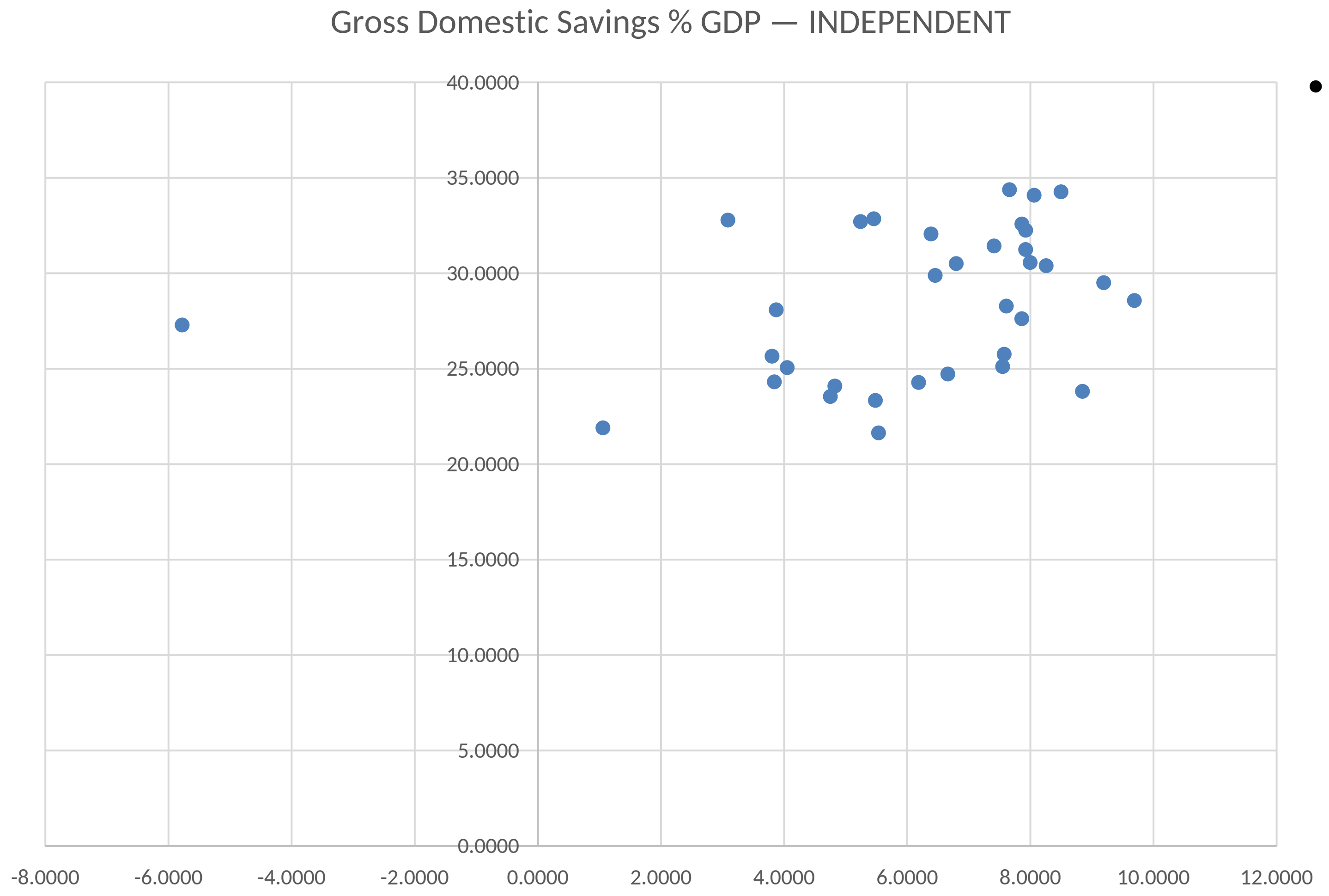
Model specification:

$$Y_t = \beta_o + (\beta_1) X_t + u_t$$

Annual movement of Gross domestic product growth rate and Gross domestic saving in india 1990–2023



- A visual comparison suggests that years with relatively higher GDS ratios are often associated with higher GDP growth.

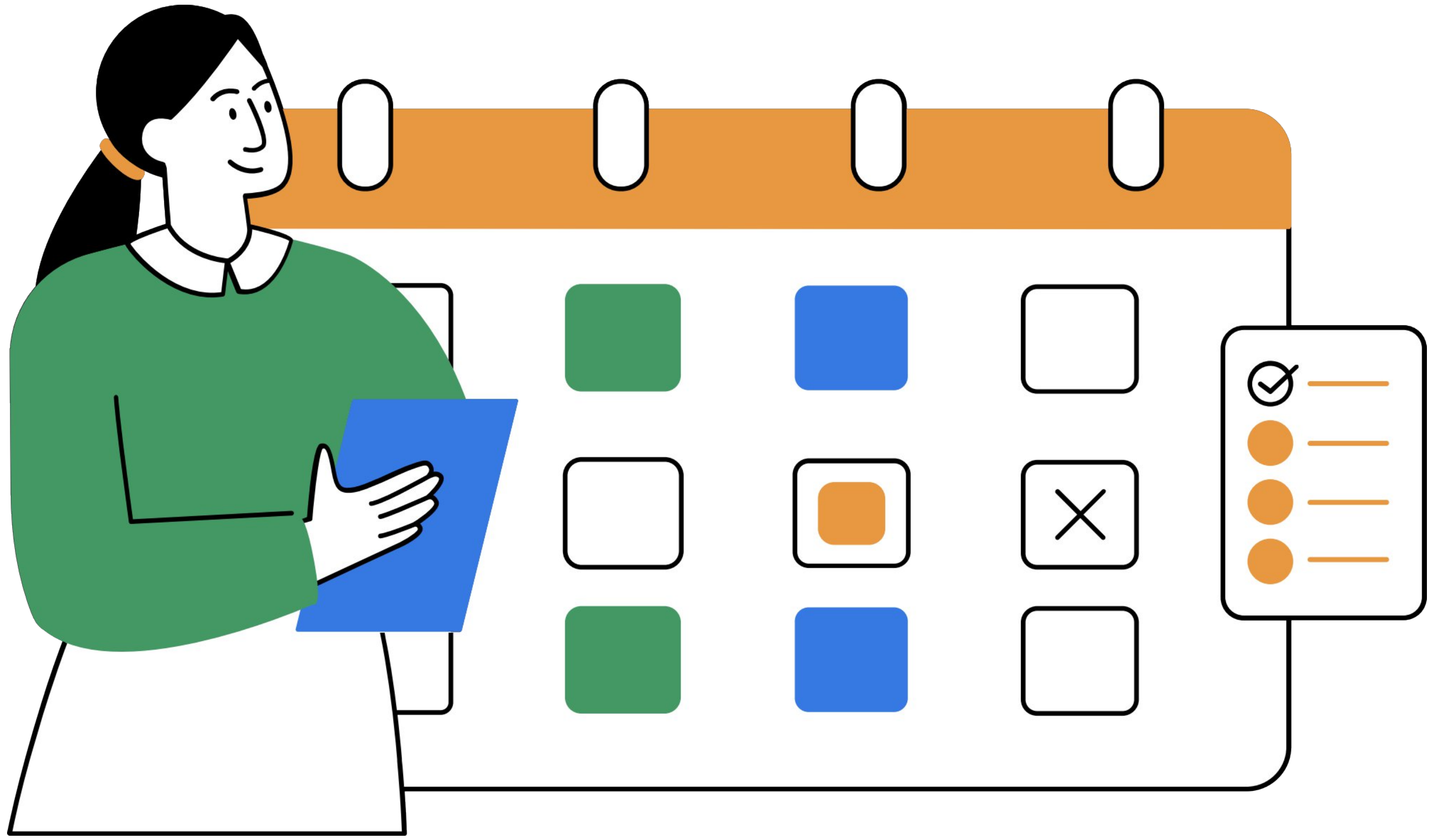
Summary Statistics

Variable	Observations	Mean	Std. Dev.	Min	Max
Time	34	6.500	9.9582	1990	2023
GDP Growth Rate (Y)	34	6.1062	2.8753	-5.7777	9.6896
GDS Percent (X)	34	28.3708	3.9001	1.6394	34.3774

- Average annual GDP growth is about 6.25 percent, but it varies considerably across years.
- The minimum GDP growth is negative, indicating at least one severe contraction year in the sample.
- The average GDS ratio is about 28.3708 percent, with values ranging from 1.63 to 34.37 percent.
- This variation in both variables is useful because regression needs enough movement in the data to detect a relationship

Regression Results

Variable	Coefficient	Std. Error	t-statistic	p-value	95% Confidence Interval
GDS Ratio (X)	0.2324	0.1237	1.8790	0.0694	[-0195, -0.4843]
Constant	-0.4871	3.5410	-0.1380	0.8914	[-7.699,-6.7256]



Model statistics:

- Number of observations = 34
- F(1,32) = 3.5300
- (Prob > F) = 0.0694
- R2 = 0.0994
- Adjusted R2 = 0.0712
- MSE = 7.6785

Interpretation:

- The coefficient of GDS is Positive , which means that higher GDS are associated with higher GDP growth rate.
- Specifically, a 1 percentage point increase in the GDS ratio is associated with about a 2.32 percentage point increase in annual GDP growth, on average.
- The p-value is 0.0694 , so the GDS coefficient is statistically insignificant at the 5 percent level.
- The model is overall insignificant because the F-test p-value is 0.0694.
- The R2 is 0.0994, so GDS alone explains about 9.94 percent of the variation in GDP growth, which suggests other macroeconomic factors also matter.

Diagnostic Test Results

Normality of errors

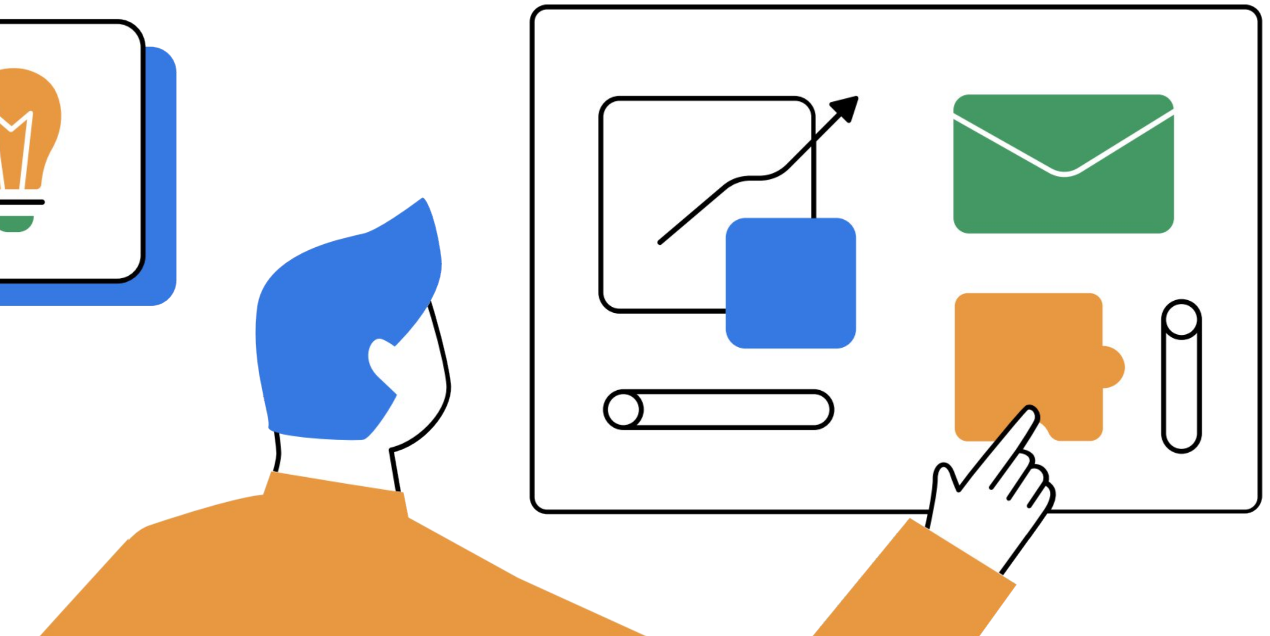
- **Skewness** = -2.342638
- **Kurtosis** = 10.870763
- **Jarque-Bera** = 118.85936
- **p-value** = 0.0000e+00
- **Conclusion:**
  Reject null hypothesis,
  Residuals are not
  normally distributed.

Heteroskedasticity

- **Breusch-Pagan test,**
  Koenker-Bassett version:
  test statistic =0.1919
- **p-value** = 0.6613
- **Conclusion:**
  Fail to reject null,
  Errors are homoskedastic
- **Cook-Weisberg version:**
  chi2(1) = 7.05
- **Prob > chi2** = 0.3304
- **Conclusion:** Fail to Reject null,
  homoskedascity present.

Autocorrelation

- **Breusch-Godfrey LM test:**
  chi2 = 0.0209,
  p-value = 0.8851
- **Runs test:**
  z = 0, **p-value** = 1
- **Durbin-Watson statistic** = 2.0123
- **Conclusion:** Fail to reject Ho,
  No evidence of autocorrelation
  at 5 percent level.



Interpretations:

- Residuals are strongly left-skewed and highly peaked, so the normality assumption is violated.
- The two versions give same evidence, where the Cook-Weisberg version points toward homoskedasdasticity. This is due to different assumptions of the tests and shows the importance of having optional tests to check for potential issues.
- There is no evidence of serial correlation in the residuals.

: Conclusion:

1. 📈 Positive but Weak RelationshipGross Domestic Savings has a statistically borderline significant positive effect on GDP Growth ($\beta_1 = +0.2324$, $p = 0.069$). A 1% rise in savings rate leads to a 0.23% increase in GDP growth — the direction is correct per economic theory, but the effect is modest.

2. **2.** 📉 **Low Explanatory Power** The model explains only **9.94% of the variation** in GDP growth ($R^2 = 0.099$, Adj. $R^2 = 0.071$). This means **~90% of GDP fluctuations** are driven by other factors not captured in this simple model — such as investment, trade, fiscal policy, or external shocks .

3. **3.** ✅ **Model is Free from Autocorrelation & Heteroscedasticity** All three autocorrelation tests (Durbin-Watson = 2.01, Breusch-Godfrey p = 0.885, Runs Test z = −0.76) and both heteroscedasticity tests (BP-KB p = 0.661, Cook-Weisberg p = 0.330) **fail to reject the null** — confirming the residuals are well-behaved on these fronts.